based on the Final Physical Prototype — Not Fabrication-Validated

because photographed breakout revisions may already contain them.

+5V_INPUT — Main 5 V input — J2

Main power switch — SW7

AMS1117-3.3 regulator module — U1

10 uF regulator input — C1

10 uF regulator output — C2

100 nF regulator bypass — C3

SIM800 4.0 V input — J3

GSM VBAT switch — SW6

1000 uF low-ESR SIM800 bulk — C4

100 nF SIM800 bypass — C5

SIM800 power header — J9

VK2828U7G5LF GPS module — J5

GPS_TX_TO_MCU_RX_PB11

GPS_RX_FROM_MCU_TX_PB10

STM32F103C8T6 Blue Pill socket — J1

BUZZER_PC13

BTN_MODE_PB12

BTN_UP_PB13

BTN_DOWN_PB14

BTN_NEXT_PB15

BTN_HELP_PA8

GSM_TX_FROM_MCU_PA9

GSM_TX_TO_MCU_RX_PA10

TEMP_DQ_PA11

ACCEL_SCL_PA6

ACCEL_SDA_PA7

MAX_SDA_PB0

MAX_SCL_PB1

OLED_SCL_PB6

OLED_SDA_PB7

10k SIM800 RX divider upper — R3

SIM800 UART / PWRKEY header — J10

47k SIM800 RX divider lower — R4

DS18B20 module — J4

4.7k DS18B20 pull-up (DNP if module fitted) — R1

ADXL345 breakout — J6

4.7k ADXL SCL pull-up (DNP) — R6

4.7k ADXL SDA pull-up (DNP) — R5

MAX30102 breakout — J7

4.7k MAX SCL pull-up (DNP) — R8

4.7k MAX SDA pull-up (DNP) — R7

0.96 inch OLED I2C module — J8

4.7k OLED SCL pull-up (DNP) — R10

4.7k OLED SDA pull-up (DNP) — R9

Mode button — SW1

Up button — SW2

Down button — SW3

Next button — SW4

Help button — SW5

1k buzzer base resistor — R2

10k buzzer base pull-down — R11

S8050 low-side buzzer driver — Q1

5 V active buzzer — BZ1

Final physical prototype governs placement; firmware governs confirmed GPIO

Not Fabrication-Validated

Independent undergraduate project reconstruction

Sheet: /

File: FallDetection_Reconstructed.kicad_sch

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